Lecture 20

Tuesday, 28 March 2023 1:29 PM

W_a → tells us uniformity of a electrochemical system

↳ $R_{ct} = \frac{\partial \eta}{\partial i}$ (depends on kinetics) ← W_{a_L}

↳ $R_e = \frac{\partial \phi_2}{\partial i}$ (essentially $\frac{\text{characteristic length}}{\text{cond.}}$) ← W_{a_T}

Low W_a → primary current dist.

Characteristics & Solutions of Primary & Secondary current dist.

↳ $\nabla i = 0$ $W_a \to 0$, the max non-uniformity in current distribution

↳ In practice, we get governing eq. and boundary condition

This is where the ref matters

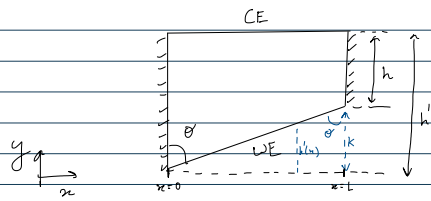
$\eta = V - \phi - \tilde{V}$

$\nabla^2 \phi = 0$; BC's: @ WE surface $\phi = V$

@ CE $\phi = 0$

This is an arbitrary constant value

Ex: Hull cell → has specific geometry to allow experimental determination of current non-uniformity



determined to see the throwing potential of electrolyte

based on how much mass accumulates @ CE at each distance

$K = \frac{L}{\tan\theta}$; $k'(x) = \frac{x}{\tan\theta}$

For primary current distribution: $\nabla^2 \phi = 0$

(Rect. coord.) $\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$

BC's: @ $y = h'$, $\phi = 0$

@ $y = K - k'$ $\phi = V$

@ $x = 0$ $\frac{\partial \phi}{\partial x} = 0$

@ $x = L$ $\frac{\partial \phi}{\partial x} = 0$

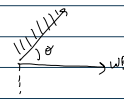
} insulating BC

$i = -\kappa \nabla \phi$

No i through an insulator ⇒ $\frac{\partial \phi}{\partial x} = 0$

Symmetry condition

Conformal mapping solution (to solve a partial diff. eq.): parameters are curve fit by realizing the property of primary current @ junction w/ insulator



$\frac{i(x)}{i_{avg}} \propto r^{\left(\frac{\pi}{2\theta} - 1\right)}$

If $\theta > \frac{\pi}{2} \to i \to \infty$

$\theta < \frac{\pi}{2} \to i \to 0$

$\theta = \frac{\pi}{2} \to i \to$ finite

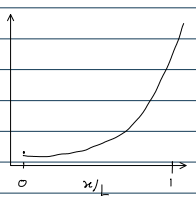
Current flowing from every where

Current line, can't flow into a narrow corner

We get:

$$\frac{i(x)}{i_{avg}} = \frac{\left(\frac{x}{L}\right)^a}{\left(1 - \frac{x}{L}\right)^b}\left(c - d\frac{K}{L}\right)$$

where a, b, c, d are constants



Secondary Current Dist.

$\nabla c_j = 0$; $Wa > 0.1$, always more uniform than primary ; kinetics are no longer negligible
(Go ϕ at electrodes should include this)

$\nabla^2 \phi = 0$;

Boundary Cond: $i = f(V, \phi)$ → BV kinetics (only describes current normal to the electrode)

↓ for uniform composition $i = -\kappa \nabla \phi$

$$i = -\kappa \left.\frac{\partial \phi}{\partial y}\right|_{y=0}$$

↳ For linear kinetics:

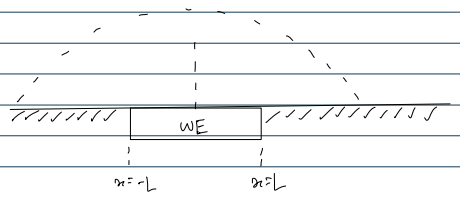
$$i = \frac{(\alpha_a + \alpha_c) i_o F}{RT} (V - \phi)$$

$$\boxed{\frac{\partial \phi}{\partial y} = \frac{-(\alpha_a + \alpha_c) i_o F}{\kappa RT} (V - \phi)}$$

BC @ WE for secondary current dist. (linear kinetics)

↳ Tafel kinetics:

ex: "Line Electrode"



Secondary current dist., linear kinetics : $\nabla^2 \phi = 0 \Rightarrow \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$

BC's: @ CE : $\phi = 0$
$x^2 + y^2 = \infty$

$$\frac{\partial \phi}{\partial y} = \frac{i_o}{\kappa} \frac{(\alpha_a + \alpha_c) F}{RT} (V - \phi)$$

Non-dimensionalizing : let $\bar{x} = \frac{x}{L}$, $\bar{y} = \frac{y}{L}$, $\bar{\phi} = \frac{\phi}{V}$

Then $\frac{\partial^2 \bar{\phi}}{\partial \bar{x}^2} + \frac{\partial^2 \bar{\phi}}{\partial \bar{y}^2} = 0$

@ $\bar{y} = 0$ $\frac{\partial \bar{\phi}}{\partial \bar{y}} = -\frac{i_o (\alpha_a + \alpha_c) F L}{\kappa RT} (1 - \bar{\phi})$, $\frac{1}{Wa_c}$

$$\boxed{\frac{\partial \bar{\phi}}{\partial \bar{y}} = \frac{-(1 - \bar{\phi})}{Wa_c}}$$ BC for linear kinetics

↳ For Tafel kinetics : $\frac{\partial \phi}{\partial y} = \frac{-i_o}{\kappa} \exp\left[\frac{\alpha_a F (V - \phi)}{RT}\right]$ @ $y = 0$

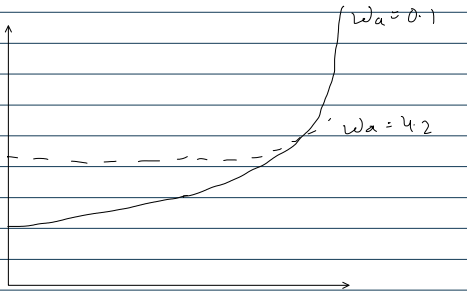
Non-dim using term in exp : $\bar{\phi} = \frac{\alpha_a F \phi}{RT}$

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$$\frac{\partial \bar{\phi}}{\partial \bar{y}} = \frac{-i_o L F}{\kappa RT} \exp\left[\frac{\alpha_a F V}{RT} - \frac{\alpha_a F \phi}{RT}\right]$$

$$\frac{\partial \bar{\phi}}{\partial \bar{y}} = -A \exp\left[-\bar{\phi}\right], \text{ where } A = \frac{i_o \alpha_a F L}{\kappa RT} \exp\left[\frac{\alpha_a F V}{RT}\right]$$

The current dist. as $W_a \to 0$ (primary



As electrode gets larger ↓ greater non-uniformity

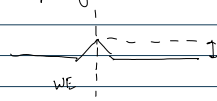
below $W_a = 0.1$ you are basically in the primary regime

We can use this understanding of W_a to get non-uniform. info

↳ eg: Dendrite formation in batteries seriously limit efficiency + development of new battery tech.

↳ uncontrolled electrode deposition

↳ Defect/asperity could create significant non-uniformity



$W_a = \frac{RT\kappa}{\alpha_c F |i_{avg}| l}$; if $i_{avg} = -50\ mA/cm^2$, $\kappa = 0.1\ S\,cm^{-1}$, $\alpha_c = 0.5$, $l = h_a$

@ $W_a = 0.1$; $h_a = 1\ cm$ ← This is high and unlikely to happen

@ $W_a = 1$ $h_a = 1\ mm$